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MULTI LINGUAL MOBILE APP FOR REAL-TIME DISEASE DETECTION

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Abstract— Our platform provides a comprehensive mobile solution for identifying cotton leaf diseases and suggesting pesticide options. It offers offline functionality for remote regions, supports local languages, and combines disease severity with the appropriate dosage for treatment. This system aims to provide farmers with a complete, accessible, and cost-effective tool for crop management. A key innovation of the system is its ability to assess disease severity using advanced image processing and machine learning algorithms, which enables the dynamic adjustment of pesticide dosage based on real-time crop health data. This targeted approach ensures effective treatment while minimizing the overuse of chemicals, promoting sustainable and environmentally friendly agricultural practices.

Keywords— Cotton disease, Mobile app, Machine learning, Image Processing, Pesticide recommendation, Disease Detection, Offline System, Smart Farming

I. INTRODUCTION

Farmers face significant challenges in identifying and controlling crop diseases, leading to major declines in crop yield. Traditional methods are often slow, inaccurate, and lack real-time guidance. Our mobile app addresses these issues by employing AI for prompt disease detection and providing precise, actionable treatment recommendations.

Rapid population growth requires higher agricultural efficiency. Traditional disease detection is manual, slow, and error-prone. AI based solutions offer faster, more accurate results.

Focus crop: Cotton and paddy

The main purpose of this research is to investigate various ways and methods for finding plant diseases early on. This is crucial since most of the existing research primarily focuses on detecting diseases in leaves and fruits. Our research wants to address this by looking at more methods to spot diseases in plants when they are still in an early stage of growth. This approach covers all areas of managing plant diseases in farming. It checks both the diseases that appear on the parts of plants we can see above the ground and those that come from the soil. The main goal of this study aims to discover better methods for early detection of plant diseases and to examine

more potential reasons why diseases occur, which makes it easier to control diseases in farming more successfully. Keeping plants healthy and getting good crop yields is very important for having enough food and staying economically stable. Plant diseases can seriously affect farming, leading to large decreases in both how much crop is produced and its quality.

Traditional ways of finding plant diseases usually have people look at the plants by hand and use their experience, which can take a lot of time, need many workers. One of the biggest changes that AI and machine learning have brought to farming is using pictures to spot plant diseases. High-quality pictures of leaves, stems, and fruits are analyzed by machine learning tools to find diseases very accurately. A strong part of machine learning, called deep learning, uses something called



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Convolutional Neural Networks (CNNs) to find patterns in images. This helps distinguish between healthy and sick with accuracy that usually surpasses human experts.

These systems not only help cut down on mistakes in diagnosis but also let farmers find diseases early, so they can act quickly and stop crops from getting lost. Besides looking at images, AI also uses data about the environment like temperature, humidity, and rainfall to predict when diseases might happen. For example, some fungi grow better in certain weather conditions. By checking real-time and past weather data, machine learning can send early warnings, helping farmers take steps to stop diseases before they spread. These predictions are especially useful in big farms where checking plants by hand is too hard. The rise of IoT devices and remote sensing has made AI systems even more powerful in farming. Sensors placed in fields and drones with cameras collect lots of data, which AI then processes to keep track of plant health and catch possible disease problems. These real-time systems are a great way to manage diseases, especially in areas with large farms or where there's not much access to expert help. The benefits of AI and machine learning in predicting plant diseases go beyond just being fast and accurate. They also help save money by cutting down on the need for field checks and lab tests.

Implementing modern maintenance techniques promotes the training of mechanics, operators, and supervisors, enhancing the technical proficiency of personnel and creating opportunities for growth. This significant figure signifies not only economic losses but also poses a threat to global food security for millions. The goal of this project is to promote collective health through occupational therapy. The proposal aims to address annual crop losses caused by plant pests. Cotton diseases are a significant concern, as noted by the TNAU Agritech Portal. The magnitude of this issue becomes even more pressing when considering future forecasts. Global demand for food, feed, fuel, and fiber is increasing, with projections indicating the world will require 50% more food by 2050 to support the growing population.

Adopting modern maintenance techniques through AI-driven disease detection offers significant benefits by enhancing the technical skills of mechanics, operators, and supervisors, there improving overall workforce proficiency and creating new opportunities for growth. The proposal aims to leverage the processing capabilities of modern smartphones to perform complex image analysis tasks.

Smartphone Penetration: In even rural areas, smartphone adoption exceeds 70% in most developing countries. Modern maintenance techniques promote the training of mechanics, operators, and supervisors, enhancing the technical level of personnel and creating opportunities.

Cost Reduction: The costs associated with AI model training and deployment have dropped by more than 80% over the last five years.

Research Gap and Innovation Focus

Adopting modern techniques for maintenance-based systems encourages the training of mechanics, operators, and supervisors, enhancing the technical proficiency of the workforce and creating opportunities for effective real-time detection systems that can operate reliably under varying lighting conditions.

This holistic approach to disease management using AI-powered mobile technology signifies not just an incremental improvement in agricultural tools, but a fundamental shift in how farmers can access, interpret, and respond to critical crop health information in real time.

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Fig.1.Cotton Leaf

Cotton Diseases

1.Cotton Leaf Curl Disease

Appearance:

Leaves curl upwards

Small growths appear beneath the leaves

The plant appears stunted

Treatment:

Apply insecticide to control whiteflies

Remove infected plants

2.Leaf Spot Disease

Appearance:

Round red spots on the leaves

The center of the spots turns white or gray

Dark rings form around the spots

Treatment:

Spray fungicide

Keep the plants dry



Fig.2:PaddyLeaf



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Paddy/Rice Diseases

1.Rice Blast

Appearance:

Diamond-shaped spots on the leaves

Spots have white to gray centers with dark borders

Leaves die and turn brown

Treatment:

Use rice varieties resistant to the disease

Apply fungicide

Manage water levels properly

2.Bacterial Blight

Appearance:

Yellow lines appear on the leaves

Water-soaked spots develop

The plant wilts

Treatment:

Apply copper-based spray

Plant early

Maintain a clean field environment

Challenges in Agriculture:

As the population grows, the challenges in agriculture are increasing rapidly. Land scarcity and water shortages are big issues in farming. Modern farming uses many technologies to solve these problems quickly.

Choosing the right seed is important.

It helps to grow better crops in the best places. Analyzing the soil, water, and pH level of the field helps in choosing the right crop for the right area, which leads to higher crop yields.

Irrigation is important for helping crops grow well.

Using machine learning helps keep the right amount of water near the plant roots. This increases the average vegetable yield. Soil moisture sensors help check how much water the plants need, and this uses less labor.

Predictive analysis helps decide when to plant, find healthy crops, and know when to add fertilizer.

Soil defect diagnosis is important.

Checking the soil helps farmers understand its condition, which is useful when growing plants.

Weather forecasting is key for stable crop growth.

The weather has a big effect on how crops grow. Productivity loss often happens because of climate issues. Using regression analysis helps predict the weather.

Detecting weeds in the field is important.

Nearly 50% of crop loss is caused by diseases, pests, and harvesting issues. Image processing helps find these problems. Classifying plants among weeds helps detect them, which helps crops grow better.

Water treatment is important because minerals are needed for plant growth. Using unsupervised analysis and detecting anomalies helps find the right amount of minerals for plant.

Objectives:

Disease Detection – The proposal aims to detect over diseases, provide accurate results with 95% precision, and deliver findings based on leaf images and AI analysis.



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Medicine Suggestions – The app will recommend the right pesticide, specify the exact amount needed, offer both cheap and expensive options, and suggest organic alternatives when possible.

Mobile App Development – The app will function without internet, be user-friendly for all farmers, support local languages, and operate on basic smartphones.

Disease Severity Assessment – The app will determine if the disease is mild, moderate, or severe, suggest treatments based on severity, calculate the exact medicine quantity, and help prevent overuse of chemicals.

Online Shopping Integration – The proposal aims to allow farmers to buy pesticides directly via the app, compare prices from different sellers, join group purchases for better deals, ensure secure payments, and offer delivery options.

Weather Integration – The app will use weather data to predict diseases, recommend the best time for spraying, provide early warnings, and help prevent diseases before they start.

Farmer Training – The goal of this project is to promote public health by utilizing occupational therapy. The proposal aims to teach disease recognition, share success stories, provide best farming practices, and connect farmers to share knowledge.

Data Collection – The goal of this project is to promote public health by tracking disease patterns, supporting researchers, aiding government policies, and improving predictions over time.

Cost Savings – The app will reduce pesticide expenses, prevent crop losses, calculate farmer savings, and show return on investment.

Environmental Protection – Modern maintenance techniques promote training for mechanics, operators, and supervisors, enhancing technical skills and supporting sustainable farming by reducing chemical pollution, protecting beneficial insects, and maintaining clean water.

System Testing – The app will be tested in various states and climates, compared with expert opinions, receive regular farmer feedback, and ensure long-term reliability.

Existing Work:

Modern disease detection systems for cotton primarily identify issues without offering treatment guidance. This leads to fragmented information for farmers, resulting in inefficient pesticide use, increased costs, and environmental damage. Most existing systems require internet access, limiting their usability in remote agricultural areas. They also fail to assess disease severity, potentially causing under- or over-treatment. The goal of this project is to promote collective health through occupational therapy. The proposal aims to integrate weather data for early disease prediction, enabling preventive actions. There is limited training or knowledge sharing to improve farmers' ability to identify diseases and adopt sustainable practices. Farmers cannot directly purchase recommended products through these systems, and important data on disease trends is not collected or analyzed to improve future accuracy.

Proposed Work:

The aim of this project is to enhance public health by means of occupational therapy. The proposal aims to promote collective wellbeing through therapeutic activities. The proposal aims to promote collective well-being through therapeutic activities.. The proposal aims to offer real-time disease detection, personalized treatment suggestions, and easy access to pesticide purchases all within a single offline-first mobile application. It tackles existing issues like the lack of treatment guidance, limited accessibility in rural regions, and the failure to capitalize on early prevention opportunities.

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Proposed System

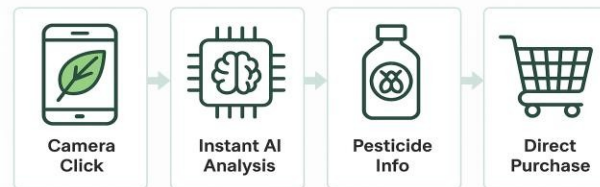


Fig.3: Proposed System

Implementing maintenance strategies based on modern techniques promotes the training of mechanics, operators, and supervisors, thereby enhancing the technical skills of the workforce and creating opportunities for improvement.

The process starts when a farmer utilizes the app's camera to capture a clear image of affected cotton or paddy leaves. The Camera Click action initiates the Instant Analysis, where deep learning models, trained on a large dataset of diseased and healthy leaf images, process the photo on the device. Advanced image preprocessing improves clarity and enables the model to assess disease severity as mild, moderate, or severe.

Once the disease is identified, the Pesticide Info stage accesses a local pesticide database to deliver precise treatment recommendations, including exact quantities, cost-effective and premium options, and organic alternatives.

This helps prevent excessive chemical usage and supports sustainable farming practices.

The final step, Direct Purchase, enables farmers to instantly buy the suggested pesticide through the app. The e-commerce module allows price comparisons between vendors, supports group buying for better deals, and offers secure payment and delivery options. All essential features remain operational offline using local storage (SQLite), with data synchronization occurring automatically when an internet connection is available.

Technologies Used

AI/Deep Learning: For real-time image-based disease classification.

Mobile App Development

Android platform

Offline support (using local storage and SQLite).

Image Processing: For preprocessing captured leaf images.

Pesticide Recommendation Engine: Based on a searchable pesticide database.

E-commerce Integration: Enables pesticide purchase.

System Architecture

The system starts by taking a photo of the plant leaf using a camera. The image is cleaned and prepared to clearly highlight key details. An AI model then checks the image and identifies if the plant has any disease. Once the disease is known, the system looks into a pesticide database to find the right treatment. The proposal aims to display information about the pesticide name, the amount to use, and the timing for application on a simple mobile interface. The proposal aims to display this information on a simple mobile app interface so that farmers can easily understand and follow it.

The system architecture is designed to detect plant diseases, identify appropriate pesticides, and provide treatment recommendations through a structured workflow. It consists of the following stages:

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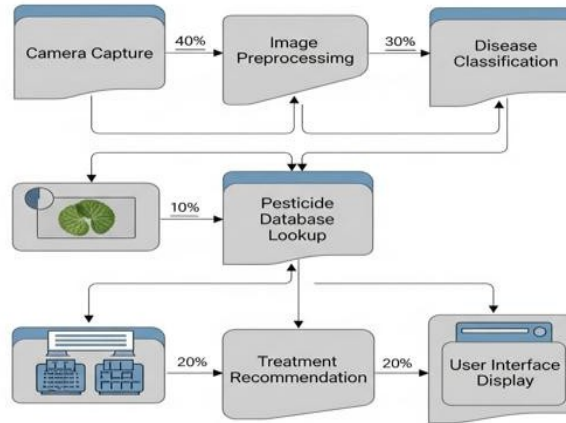


Fig.4: System Architecture

Step-1: Camera Capture:

Camera capturing high-resolution images of plant leaves using a camera. This step ensures the collection of sufficient visual data to accurately identify disease symptoms. The captured images are then sent for further processing.

Step-2: Image Preprocessing:

This process may involve resizing, improving contrast, removing the background, and normalizing the images. The images to improve their quality and remove noise. Preprocessing helps retain only the relevant features of the leaf, thereby improving the accuracy of the disease classification model.

Step-3: Disease Classification:

The model examines patterns, colors, and textures to match them with known disease categories. The classification output is then used to determine the appropriate pesticide.

Step-4: Pesticide Database Lookup:

Once the disease is classified, the system consults a pesticide database that contains detailed information about suitable pesticides for each disease. This database serves as a reference to ensure that the recommendations are accurate and current.

Step-5: Treatment Recommendation:

Based on the database results, the system provides treatment recommendations, including the name of the pesticide, dosage, and application instructions. It ensures that farmers or users receive clear guidance to effectively address the disease.

Step-6: User Interface Display:

The final stage presents the classification results, pesticide details, and treatment recommendations via an easy-to-use interface. The UI is designed to be user-friendly, allowing even non-technical users to access and understand the information quickly.

This architecture ensures a smooth flow from image capture to actionable output, combining computer vision, database integration, and user-centric design.

II OBSERVATIONS OF SUMMARY

Table-1: Survey on Target Crop Diseases and Treatment Protocols

Crop	Disease	Visual Symptoms	Primary Treatment	Innovation
Cotton	Cotton leaf spot	Red spots with white centers	Use of fungicide	Calculate treatment precisely



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Cotton	Cotton leaf curl	Upward leaf curl, stunted plant	Remove infected plants	Dynamic severity- based dosage
Paddy	Rice blast	Diamond- shaped spots, dark edges	Manage fungicide and water	Combine with environment al factors
Paddy	Bacteri alblight	Yellow lines, wilting	Copper- based spray, early planting	Preventive approach emphasis

Table-2: Technology Implementation Framework

Technology Component	Implementation approach	Innovation Aspect	Technical advantage
AI/Deep Learning	Instant image analysis	No need for internet	Processing done on the device
Mobileapp development	Android platform	Wide device compatibility	Basic smart phone support
Offline support	Sqlite local storage	Complete functionality	Rural area accessibility
Image processing	Advanced preprocessing	Quality enhancement	Mobile optimization
Smart recommendation	Easy-to- search list of Pesticides	Tailored suggestions	Accurate application
Online shopping made easy	Secure ways to pay	Fast and direct purchasing	One-step solution

Table-3: Expected Impact Observations

ImpactArea	Metrics	Expected outcome
Economic	Cost Savings	Lower cost for pesticides, avoided damage to crops
Environmental	Chemical usage	Reduced chemical pollution, protected beneficial insects
Technical	Skill development	Enhanced technical proficiency of agricultural workers
Accessibility	Rural areas	Get better access to expert knowledge that they might not easily find
Sustainability	Farming methods	Good for the environment and encourage eco- friendly agriculture
Food Security	Crop Yield	Increase agricultural productivity

III. CONCLUSION

Our multilingual mobile app provides a complete solution for realtime disease detection and pesticide management in farming. It uses artificial intelligence, works even without an internet connection, and includes an online store, helping farmers grow more crops, save money, and use farming methods that are better for the environment. By using deep learning, the app has made plant disease detection much better, faster, and more reliable. It also connects research from universities and labs with what farmers actually do on the ground, so cutting-edge AI tools can be used by people who work the land every day. This app helps farmers in many ways, like saving money, preventing plant diseases, and making it easier to use new technology. It also helps the plant by using less pesticide, keeping soil healthy, protecting helpful insects, and supporting farming that



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works with nature. In the future, the app could grow to include tools that predict problems before they happen, connect with smart farming devices, and help manage crops more precisely. This makes the app flexible and useful in many areas and for different types of crops. It's a simple, practical AI tool that helps make farming smarter and more sustainable

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